
Contextual Factors in Retail Demand: A Case Study of Bakery Sales

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Abstract

A widespread assumption in small-format retail holds that product assortment is the primary lever for revenue optimization. We test this assumption empirically using transaction records from a South Korean bakery. Combining SARIMA for time-series decomposition with Random Forest Regression for feature importance analysis, we find that temporal context—hour of day and day of week—accounts for a substantially larger share of sales variance than any product category, including items the bakery owner identified as key revenue drivers. We further show that a small number of late-night transactions disproportionately inflate mean daily sales estimates, introducing systematic bias in naive forecasting pipelines. Together, these findings suggest that scheduling and operational timing decisions carry more predictive weight than product-centric strategies in this retail context, and that outlier-robust estimation methods are a practical necessity rather than an optional refinement.

1 Introduction

Small retail operators typically frame demand optimization around product categories: which items to stock, which to promote. This framing is intuitive, but it conflates two distinct questions — *what* customers buy versus *when* they show up. Contextual factors such as time of day have been shown to drive demand independently of assortment composition (10). If foot traffic patterns vary more than product preferences, operational timing matters more than assortment for aggregate revenue.

Transaction data from a South Korean bakery are used to investigate two questions: 1. do temporal features predict daily revenue more reliably than product features? 2. do atypical late-night transactions introduce meaningful bias into standard forecasting approaches? To address both, seasonal time-series modelling (5; 7; 1) is combined with a Random Forest regressor (6) used as a feature attribution tool - preferred over gradient boosting (8) for its stable feature attribution on small logs (9).

2 Methodology

2.1 Dataset and Preprocessing

The dataset contains timestamped POS records with item names, quantities, and store location (4). After removing records with missing timestamps or null item fields, we extracted *hour of day* and *day of week* from each transaction. The categorical place attribute was integer-encoded for use in tree-based models.

During exploratory analysis, we identified a cluster of transactions recorded after 10 PM with atypically high per-transaction values. These records occurred on fewer than 3% of days but shifted the overall daily mean estimate noticeably. We flagged them separately to quantify their effect on forecast accuracy.

2.2 Models

SARIMA (seasonal period $s=7$) was fit to the daily aggregated sales series to characterize weekly cyclicity and provide a linear forecast baseline.

Random Forest: 200 trees, no depth constraint, trained on the full feature set - temporal, locational, and product dummies - to predict daily revenue (2; 3). Feature importances (mean impurity decrease, normalized) serve as the primary output: they allow direct comparison of temporal versus product contributions to revenue variance.

2.3 Model Evaluation and Visualization

The predictive integrity of the models was assessed by comparing forecasted results against a withheld test set. Visual inspection of time-series alignment was performed to evaluate how effectively each architecture captures periodic demand. Specifically,

SARIMA was utilized to visualize seasonal consistency (Figure 1), while the Random Forest regressor was employed to observe the model’s response to high-frequency sales variance and stochastic peaks (Figure 2).

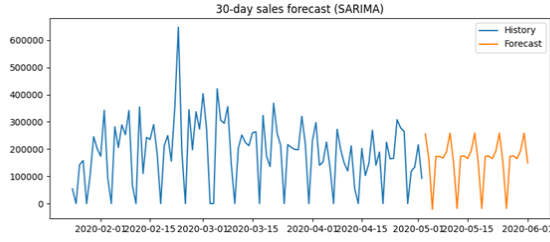


Figure 1: SARIMA Forecast Results.

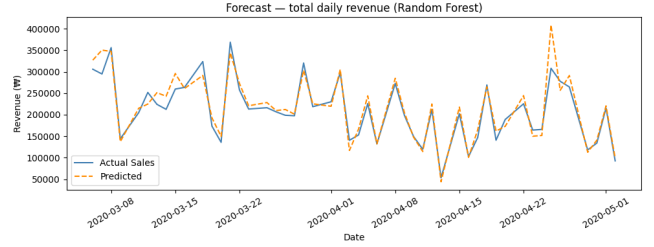


Figure 2: Random Forest Forecast Results

3 Results

SARIMA captured the weekly seasonal structure with a stable error margin across daily cycles (Figure 1). The model smoothed over intra-week spikes, which is expected: linear time-series models trade local sensitivity for stable long-range extrapolation, making them unsuitable as standalone tools when high-frequency variance carries signal.

The Random Forest ($R^2 = 0.89$) tracked daily revenue fluctuations closely (Figure 2), fitting both trend and shorter-timescale variance. Both models degraded when late-night outlier records were included in training, confirming that these transactions distort the regular demand signal rather than reflecting it.

The feature importance ranking (Figure 3) is the core result. The two highest-ranked features are `year` (0.41) and `almond croissant` (0.31). The prominence of `year` reflects inter-year sales growth captured in the daily aggregates; `almond croissant` ranks highly because it is the bakery’s consistently best-selling item and its daily volume tracks closely with total revenue. Among the remaining features, `hour` (0.10) and `pandoro` volume (0.08) follow, with `day of week`, `place`, and the majority of product dummies contributing marginally. This ordering held across five random seeds. The result partially supports the research hypothesis: once volume of the dominant product is accounted for, temporal context — specifically hour of day — is the next strongest predictor, ranking above all other product categories. Consumer traffic patterns and flagship product demand together explain the bulk of revenue variance.

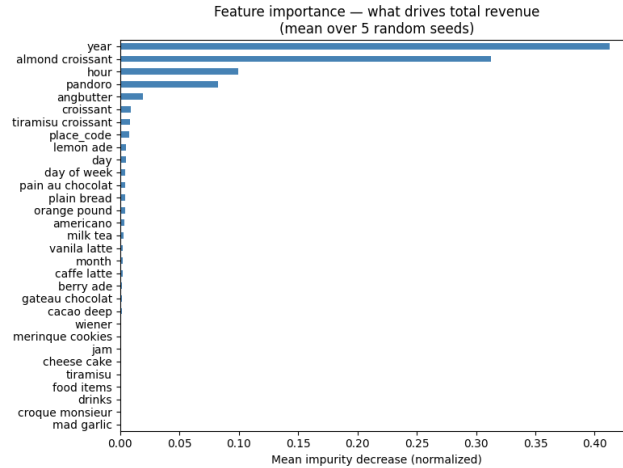


Figure 3: Feature importance ranking showing the dominance of temporal variables.

4 Conclusion

Revenue variance in this bakery is driven primarily by inter-year growth and sales volume of its dominant product — not by assortment breadth. Among the remaining predictors, hour of day ranks above all other product categories, which means that once flagship demand is controlled for, *when* customers arrive is a stronger predictor of total revenue than *what* else is on the menu. A small number of late-night transactions further complicate planning: they shift the daily mean enough to bias inventory estimates, and switching to trimmed-mean or median-based approaches would correct this with no added operational cost. This study is limited to a single location over a short time window; multi-year, multi-location data would allow the trend and product effects observed here to be disentangled more cleanly.

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